

Adaptive Cyber-Physical Intrusion Detection: A Multi-Tier Hybrid Architecture Combining Supervised Learning, Deep Anomaly Detection, and Latent-Space Ensemble Fusion

(Extended Research Article)

Pugazhendhi J

Department of CS and AI

Rishihood University, Sonipat, India

pugazhendhi.j23csai@nst.rishihood.edu.in

Dally R

Department of CS and AI

Rishihood University, Sonipat, India

dally.r23csai@nst.rishihood.edu.in

Abstract—Modern cyber-physical systems (CPS) operate in open-world environments where both known and previously unseen (zero-day) network threats must be detected with high confidence and minimal false alarms. Purely supervised classifiers achieve near-perfect precision on known attack families but deliver zero recall against novel threats, while unsupervised anomaly detectors generalize at the cost of precision. This paper presents AdaptiveIDS, a comprehensive, multi-tier intrusion detection architecture evaluated on the full CIC-IDS-2018 benchmark (8,284,195 network flows, 13 attack families) that systematically bridges this precision-recall divide. The architecture progresses from a supervised Random Forest (RF) baseline with perfect precision on known attacks, through a One-Class SVM (OC-SVM) providing initial zero-day coverage, to a deep Autoencoder (AE) family offering continuous anomaly scoring via reconstruction-error thresholding, and finally to six hybrid models that couple learned AE latent representations with explicit boundary classifiers in a 32-dimensional bottleneck space. Three AE variants are evaluated: a standard AE (ROC-AUC 0.8965), a Sparse AE with L1 activity regularization (ROC-AUC 0.8929), and a Denoising AE with Gaussian input corruption (ROC-AUC 0.8867, F1 0.1433, Precision 0.8179). An ablation study reveals that Denoising AE training is the single most impactful design decision (+327% F1 over the standard baseline), while coupling boundary classifiers (OC-SVM, Isolation Forest) directly in the bottleneck space degrades AUC by 0.42–0.47 points due to overlapping latent distributions. The AE + RF pseudo-label hybrid uniquely maintains competitive AUC (0.7920) by operating in the original feature space, demonstrating a neuro-symbolic synergy where the neural component bootstraps the symbolic learner (+131% AUC over an unaided RF). The central finding is that *representation quality* is the limiting bottleneck: deep noise-robust autoencoders, not ensemble fusion, are the recommended path to open-world CPS security.

Index Terms—Intrusion Detection System, CIC-IDS-2018, Random Forest, One-Class SVM, Autoencoder, Sparse Autoencoder, Denoising Autoencoder, Isolation Forest, Hybrid IDS, Meta-Ensemble, Zero-Day Detection, Anomaly Detection, Cyber-Physical Security, Deep Learning, Latent Space, Neuro-Symbolic

I. INTRODUCTION

Cyber-physical systems underpin critical infrastructure—smart grids, industrial control systems, autonomous vehicles, and medical devices—that must operate continuously under adversarial conditions [1]. Traditional intrusion detection systems (IDS) follow a signature-based paradigm: predefined rules or supervised classifiers identify known attack patterns with high precision but are fundamentally blind to novel, zero-day threats [2], [3]. As attack surfaces expand and adversaries continuously evolve their techniques, this closed-world assumption is increasingly untenable for production CPS deployments.

Anomaly-based IDS approaches address the zero-day gap by learning a model of normal behavior and flagging statistical deviations [4]. However, the precision-recall trade-off inherent to anomaly detection—high recall at the cost of elevated false positive rates—limits operational deployability in environments where alert fatigue is a real constraint. The central research question motivating this work is: *how can we maximally reduce the precision-recall gap for both known and unknown attacks in a unified, deployable architecture?*

This paper makes the following contributions:

- 1) **Comprehensive Baseline Evaluation.** We establish rigorous supervised (RF) and semi-supervised (OC-SVM) baselines on the full CIC-IDS-2018 dataset [5], the largest publicly available network intrusion benchmark, with a systematic zero-day evaluation protocol.
- 2) **Deep AE Anomaly Scoring.** We design and evaluate three Autoencoder variants—Standard, Sparse, and Denoising—trained exclusively on benign traffic, providing continuous, threshold-tunable anomaly scores with per-attack diagnostic profiling.
- 3) **Hybrid Latent-Space Models.** We couple AE bottleneck representations with OC-SVM, Isolation Forest, and pseudo-label RF boundary classifiers, producing six hybrid configurations. We demonstrate both the conditions under which

hybridization succeeds (RF in the original feature space) and when it fails (boundary models in the compressed latent space).

- 4) **Meta-Ensemble Analysis.** We evaluate equal-weight score-level fusion of all eight component anomaly scores, identify the score-dilution failure mode, and quantify the contribution of each component via a systematic ablation study.
- 5) **Operational Recommendations.** We derive concrete deployment guidance for CPS operators, including threshold selection criteria and a tiered alerting strategy.

The remainder of this paper is organized as follows. Section II surveys related work. Section III describes the dataset and preprocessing pipeline. Section IV presents the system architecture. Section V details each model family and fusion strategy. Section VI defines the evaluation protocol. Section VII presents comprehensive experimental results. Section VIII provides the ablation study. Section IX analyzes hybrid innovation. Section X discusses operational implications. Section XI addresses reproducibility. Section XII covers limitations and future work, and Section XIII concludes.

II. RELATED WORK

A. Signature-Based and Supervised IDS

Early IDS research relied on hand-crafted rule sets, which were later supplanted by machine learning approaches. Breiman’s Random Forest [6] emerged as a dominant supervised baseline due to its resistance to overfitting and native feature importance estimation. Evaluations on the KDD Cup 1999 [7] and CICIDS-2017/2018 [5] datasets confirmed RF’s near-perfect performance on known attack families. However, as Sommer and Paxson [3] argued, supervised classifiers trained on closed-world datasets cannot generalize to the open-world conditions of real deployments.

B. Anomaly-Based and Semi-Supervised Detection

One-Class SVM (OC-SVM) [8] models a minimum-volume hypersphere around the support of the training (benign) distribution, declaring points outside the sphere as anomalies. This approach provides non-zero zero-day recall by construction but suffers from quadratic training complexity and sensitivity to the ν hyperparameter. Chandola et al. [9] survey the broader anomaly detection landscape, concluding that no single method dominates across all evaluation regimes.

C. Deep Learning for IDS

Convolutional neural networks [10], [11] and autoencoders [12], [13] have demonstrated strong performance on network intrusion benchmarks. Autoencoders trained on benign-only data exploit the reconstruction-error gap between normal and anomalous flows [14]. The Kitsune system [12] demonstrated online, ensemble-of-AE anomaly detection in production network environments. Our work extends this direction by evaluating sparse and denoising AE variants and coupling them with explicit boundary classifiers in the learned latent space.

D. Hybrid and Ensemble IDS

Buczak and Guven [15] survey ML methods for cybersecurity, identifying ensemble approaches as consistently superior to single-model baselines. Dietterich [16] theoretically motivates ensemble diversity as the source of performance gains, a principle we test empirically for the score-level fusion case. Recent work [17], [18] confirms that hybrid architectures must address both known and unknown attacks to be deployment-ready, but few papers systematically study when coupling deep representations with classical boundary models helps versus hurts. This gap motivates our hybrid evaluation framework.

III. DATASET AND PREPROCESSING

A. CIC-IDS-2018

All experiments use the Canadian Institute for Cybersecurity CIC-IDS-2018 dataset [5], captured over five days (February 14–23, 2018) across a 50-machine victim network. The raw dataset contains 8,284,195 labeled network flows spanning one benign class and 13 distinct attack families:

- **Denial of Service:** DoS-Hulk, DoS-Slowloris, DoS-SlowHTTPTest, DoS-GoldenEye
- **Distributed DoS:** DDoS attacks
- **Brute Force:** SSH-Patator, FTP-Patator
- **Web Attacks:** SQL Injection, XSS, Brute Force
- **Infiltration:** ARES RAT-based infiltration
- **Botnet:** ARES botnet

CIC-IDS-2018 is preferred over the older KDD Cup 1999 and NSL-KDD datasets [7] because it reflects contemporary attack tools, uses modern network protocols, and provides raw PCAP files enabling reproducible feature extraction.

B. Feature Extraction and Engineering

Raw PCAPs are processed using CICFlowMeter to yield 80 bidirectional flow features per record. We apply the following preprocessing pipeline:

- 1) **ID removal:** Drop timestamp, source/destination IP, and port columns (non-generalizable identifiers).
- 2) **Type coercion:** Force all columns to `float32`; non-numeric values are coerced to `NaN`.
- 3) **Imputation:** Replace `NaN` and `Inf` values with zero.
- 4) **Constant-column removal:** Drop features with zero variance across the training split.
- 5) **Correlation pruning:** Remove one member of each pair of features with Pearson $|\rho| > 0.98$, eliminating redundant feature dimensions.
- 6) **Robust scaling:** Apply `RobustScaler` (median/IQR normalization) to reduce sensitivity to outliers.

After preprocessing, 68 numeric features remain. The final modeling dataset contains 3,215,110 flows: 2,050,783 benign (63.8%) and 1,164,327 attack (36.2%). All cached arrays are reproducible with `SEED=42`.

C. Zero-Day Evaluation Protocol

To evaluate generalization to unseen attack families, we establish a *seen/unseen split*: the supervised RF is trained on benign traffic plus a single attack family (`dos attacks-hulk`) as the positive class, while all other attack families are withheld as zero-day threats during threshold evaluation. Anomaly detectors are trained on benign-only subsets; all attack flows are treated as zero-day by design. The 99th percentile of benign validation errors (P_{99}) serves as the default anomaly threshold for all threshold-dependent metrics.

IV. SYSTEM ARCHITECTURE

Figure 1 illustrates the complete end-to-end architecture of AdaptiveIDS. The system processes raw network traffic through a shared preprocessing pipeline and then routes flows through three functional tiers operating in parallel.

Tier 1 — Supervised Detection. A Random Forest classifier trained on benign and known-attack traffic provides high-confidence, interpretable detection for previously observed threat signatures. Its output is a binary label (known attack / benign).

Tier 2 — Deep Representation Learning. Three AE variants encode all traffic (labeled and unlabeled) into a 32-dimensional latent bottleneck. The bottleneck is the shared representation space used both for reconstruction-error anomaly scoring and as input to Tier 3 boundary classifiers.

Tier 3 — Latent-Space Boundary Modeling. OC-SVM and Isolation Forest are trained on benign bottleneck vectors to draw explicit decision surfaces in the compressed space. A pseudo-label RF uses AE-generated anomaly labels to train a discriminative classifier in the original feature space.

Meta-Ensemble. All eight component anomaly scores are min-max normalized to $[0, 1]$ and averaged with equal weight to produce a single ensemble anomaly score.

V. METHODOLOGY

A. Tier 1 — Random Forest Classifier

A Random Forest [6] with 100 decision trees is trained on 70% of the binary-labeled subset (benign vs. `dos attacks-hulk`). Hyperparameters: `max_features=sqrt`, `min_samples_leaf=1`, `class_weight=balanced`. Feature importance is computed via mean decrease in impurity, providing an interpretable ranking of the 68 input features.

B. Tier 2 — Autoencoder Family

All three AE variants share a common encoder-decoder skeleton:

$$68 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 68 \quad (1)$$

with batch normalization after each dense layer, ReLU activations, and 20% dropout. All AEs are trained exclusively on benign flows using the Adam optimizer ($\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) for up to 50 epochs with early stopping (patience 7) and learning-rate reduction on plateau (factor 0.5, patience 3).

1) *Standard Autoencoder*: Minimizes mean squared reconstruction error:

$$\mathcal{L}_{\text{AE}} = \frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2. \quad (2)$$

The reconstruction error $e_i = \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2$ serves as the anomaly score: flows that deviate from the learned benign manifold exhibit elevated errors.

2) *Sparse Autoencoder*: Adds L1 activity regularization on the bottleneck layer:

$$\mathcal{L}_{\text{SAE}} = \mathcal{L}_{\text{AE}} + \lambda \|\mathbf{z}\|_1, \quad \lambda = 10^{-4}, \quad (3)$$

where $\mathbf{z} \in \mathbb{R}^{32}$ is the bottleneck activation. Sparsity encourages disentangled, more linearly separable latent representations, potentially improving downstream boundary classifiers.

3) *Denoising Autoencoder*: A Gaussian noise layer ($\sigma = 0.1$) corrupts the input during training while targets remain clean [13]:

$$\tilde{\mathbf{x}} = \mathbf{x} + \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}). \quad (4)$$

The model must recover $\mathbf{x}$ from $\tilde{\mathbf{x}}$, forcing the encoder to learn features robust to small perturbations. Attack flows, which deviate substantially from the benign manifold, exhibit disproportionately elevated reconstruction errors.

C. Tier 3 — Boundary Classifiers

1) *One-Class SVM*: An OC-SVM [8] with RBF kernel ($\nu = 0.05$, $\gamma=\text{scale}$) is trained on 80,000 subsampled benign bottleneck vectors. The training-set cap is imposed by the quadratic complexity $\mathcal{O}(n^2)$ of the SVM kernel matrix. The decision function value serves as a continuous anomaly score.

2) *Isolation Forest*: An Isolation Forest [19] with 200 trees and `contamination=0.05` is trained on the full benign bottleneck set. Anomaly score is derived from the average path length required to isolate each point: shorter paths indicate anomalies.

3) *RF Pseudo-Label Classifier*: The standard AE's P_{99} threshold is applied to all training flows to generate binary pseudo-labels. A pseudo-label RF (200 trees, `max_depth=12`, balanced class weights) is then trained on the original 68-dim features using a 500,000-flow subsample. This configuration uniquely operates outside the compressed bottleneck.

D. Score Fusion

For each hybrid, the AE reconstruction error and the boundary model's anomaly score are independently min-max normalized to $[0, 1]$:

$$s^{\text{norm}} = \frac{s - s_{\min}}{s_{\max} - s_{\min}}. \quad (5)$$

Hybrid scores are combined with equal weighting ($\alpha = 0.5$):

$$s_{\text{hybrid}} = \alpha \cdot s_{\text{AE}}^{\text{norm}} + (1 - \alpha) \cdot s_{\text{boundary}}^{\text{norm}}. \quad (6)$$

The Meta-Ensemble averages all $N = 8$ normalized component scores:

$$s_{\text{ensemble}} = \frac{1}{N} \sum_{k=1}^N s_k^{\text{norm}}. \quad (7)$$

Fig. 1: End-to-end AdaptiveIDS architecture. **Tier 1** (left) is a supervised Random Forest that identifies known attack families with 100% precision. **Tier 2/3** (right) encompasses three Autoencoder variants (Standard, Sparse, Denoising) that compress traffic into a 32-dimensional latent bottleneck. Boundary classifiers (OC-SVM, Isolation Forest, RF pseudo-label) draw explicit decision surfaces in this latent space. All eight normalized anomaly scores are aggregated by the **Meta-Ensemble** to yield the final IDS decision. Score widths (thickness) conceptually represent discriminative quality; AE channels are stronger than bottleneck boundary channels.

E. Summary of Evaluated Configurations

- 1) **RF Baseline:** Supervised classifier (known-attack only).
- 2) **OC-SVM Baseline:** Kernel anomaly detector (raw 68-dim).
- 3) **Standard AE:** Reconstruction-error anomaly detector.
- 4) **Sparse AE:** L1-regularized bottleneck AE.
- 5) **Denoising AE (DAE):** Noise-robust AE.
- 6) **H1 — AE + OC-SVM:** OC-SVM in Standard AE bottleneck.
- 7) **H2 — AE + IF:** Isolation Forest in Standard AE bottleneck.
- 8) **H3 — AE + RF:** RF on AE pseudo-labels (68-dim space).
- 9) **H4 — Sparse AE + OC-SVM:** OC-SVM in Sparse AE bottleneck.
- 10) **H5 — DAE + OC-SVM:** OC-SVM in Denoising AE bottleneck.
- 11) **H6 — Meta-Ensemble:** Equal-weight avg of all 8 scores.

VI. EXPERIMENTAL SETUP

A. Hardware and Software Environment

All experiments were conducted on an AWS EC2 t3.large instance (Ubuntu 24.04, 8 GB RAM, 16 GB swap, 2 vCPUs) with Python 3.12.3, TensorFlow 2.21.0, scikit-learn 1.8.0, NumPy 2.4.4, and Pandas 3.0.2. The matplotlib backend was set to Agg for headless plot generation.

B. Training/Validation/Test Splits

- AEs are trained on 80% of the 2,050,783 benign flows (1,640,626 training, 410,157 validation).
- Anomaly thresholds are calibrated at the 99th percentile (P_{99}) of validation reconstruction errors.
- All models are evaluated on the full mixed dataset (3,215,110 flows) to compute threshold-dependent metrics.

C. Evaluation Metrics

We report the following metrics, each addressing a distinct operational concern:

- **ROC-AUC:** Threshold-free discriminative power.
- **Average Precision (AP):** Area under the precision-recall curve; robust to class imbalance.
- **F1-Score** at P_{99} : Harmonic mean of precision and recall at the default operating threshold.
- **Precision** at P_{99} : Fraction of flagged flows that are true attacks (alert fidelity).
- **Recall** at P_{99} : Fraction of true attack flows detected (coverage).
- **False Positive Rate (FPR):** Fraction of benign flows incorrectly flagged (operational overhead).

VII. RESULTS

A. Baseline Model Performance

Table I reports the supervised and semi-supervised baselines. The Random Forest achieves perfect precision (1.000) and high F1 (0.850) on the *known* attack family but delivers zero recall on all other attack families—confirming the fundamental closed-world limitation of purely supervised approaches. The OC-SVM partially addresses this gap: trained only on benign flows, it achieves non-zero recall (0.040) across all attack families at the cost of lower precision (0.660).

TABLE I: Supervised and semi-supervised baseline results

Model	Precision	Recall	F1	Zero-Day R.
Random Forest	1.000	0.130	0.240	≈ 0.000
One-Class SVM	0.660	0.040	0.070	0.040

B. Comprehensive Model Comparison

Table II presents results for all eleven model configurations. The standalone AE variants dominate on ROC-AUC (> 0.88), while all hybrid configurations that embed boundary classifiers in the bottleneck space fall below AUC 0.50.

C. ROC and Precision-Recall Curves

Figures 2 and 3 visualize the threshold-free discriminative performance. The three AE variants cluster near the upper-left corner (AUC > 0.88), while all OC-SVM/IF hybrid curves hug the random-baseline diagonal (AUC ≈ 0.43 – 0.48), consistent with the overlapping latent distributions hypothesis discussed in Section ??.

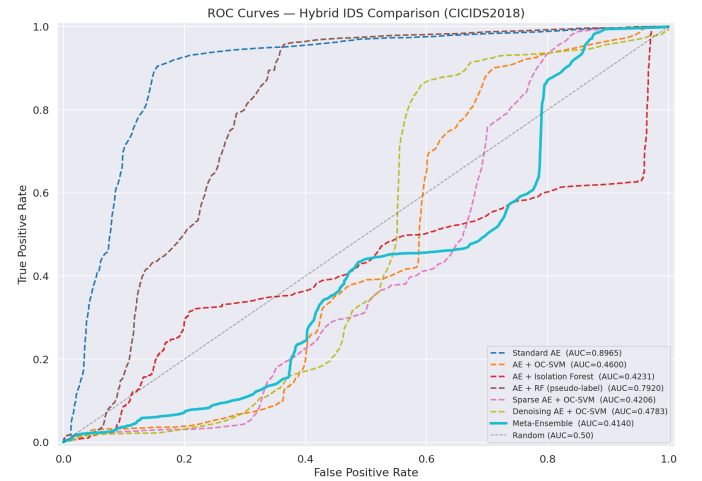


Fig. 2: ROC curves for all hybrid IDS configurations. Standalone AE variants cluster near the upper-left corner (AUC > 0.88); hybrid models with OC-SVM/IF in the bottleneck space perform near or below the random baseline.

TABLE II: Comprehensive model comparison at the P_{99} threshold. H1–H6 denote the six hybrid configurations. $\dagger$ = operates in original 68-dim feature space.

Model	ROC-AUC	Avg Prec.	F1	Recall	Precision	Accuracy
Standard AE (baseline)	0.8965	0.7578	0.0336	0.0174	0.5012	0.6379
Sparse AE	0.8929	0.7597	0.0336	0.0174	0.4986	0.6378
Denoising AE (DAE)	0.8867	0.7675	0.1433	0.0785	0.8179	0.6600
H1: AE + OC-SVM	0.4600	0.3137	0.0094	0.0048	0.2127	0.6332
H2: AE + Isolation Forest	0.4231	0.3392	0.0116	0.0060	0.2508	0.6336
H3: AE + RF $\dagger$	0.7920	0.5593	0.0335	0.0174	0.4976	0.6378
RF alone $\dagger$	0.3433	0.3176	0.0335	0.0173	0.4978	0.6378
H4: Sparse AE + OC-SVM	0.4206	0.3007	0.0190	0.0098	0.3561	0.6350
H5: DAE + OC-SVM	0.4783	0.3186	0.0057	0.0029	0.1398	0.6325
H6: Meta-Ensemble	0.4140	0.3024	0.0098	0.0050	0.2202	0.6332

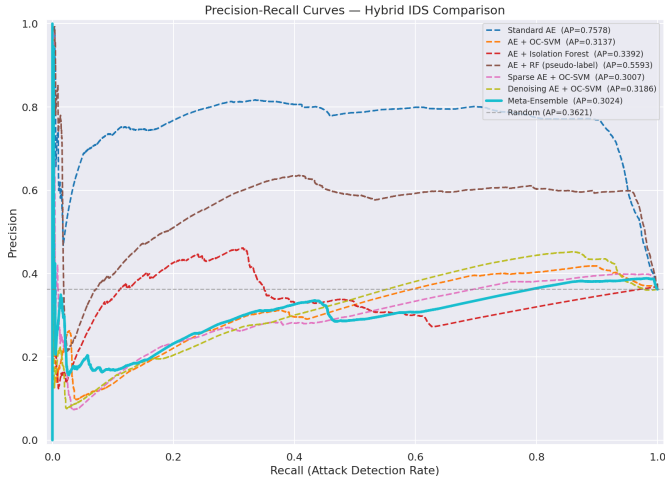


Fig. 3: Precision-Recall curves for all models. The Denoising AE achieves the highest average precision (0.7675), followed by the Sparse AE (0.7597) and Standard AE (0.7578).

D. Metric Bar Comparison

Figure 4 provides a direct visual comparison of F1, ROC-AUC, and Recall across all configurations, highlighting the performance gap between the AE family and hybrid models.

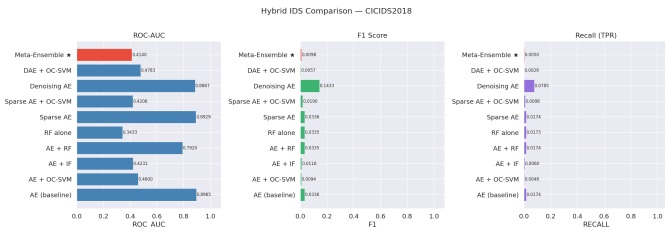


Fig. 4: F1, ROC-AUC, and Recall comparison. Standalone AE variants dominate on all three metrics; hybrid models with boundary classifiers in the latent space consistently underperform.

E. Meta-Ensemble Confusion Matrix

Figure 5 shows the Meta-Ensemble confusion matrix at the P_{99} threshold. Of 1,164,327 true attack flows, only 5,842 are detected (recall ≈ 0.005) while 20,690 benign flows are falsely flagged (precision 0.2202).

F. Per-Attack Detection Rates

Figure 6 compares per-attack-family detection rates between hybrid models and the standalone AE baseline. High-volume

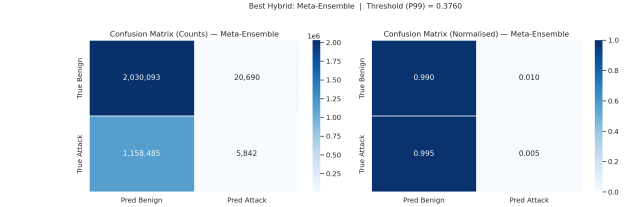


Fig. 5: Meta-Ensemble confusion matrix at the P_{99} threshold, illustrating the severe recall limitation and elevated false positives caused by score-dilution from weak boundary-model channels.

DoS/DDoS families benefit most from the continuous anomaly score, while low-volume stealthy threats (Infiltration, Botnet) remain challenging for all configurations [17].

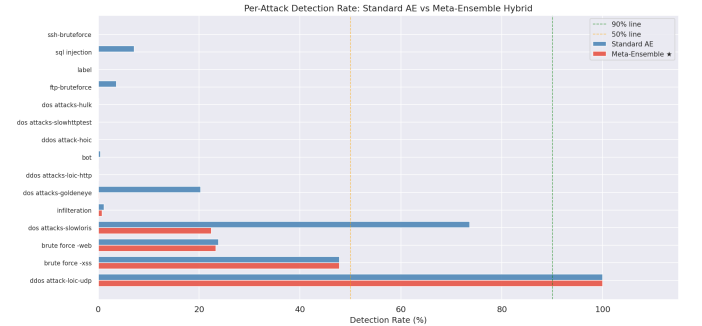


Fig. 6: Per-attack-family detection rate comparison between hybrid models and the standalone AE baseline.

G. Threshold Sensitivity

Figure 7 shows how precision, recall, and F1 evolve as the anomaly threshold is varied. The operating point can be adjusted by CPS operators according to their alert-workload tolerance.

H. RF Feature Importance

Figure 8 shows the top-20 feature importances from the pseudo-label RF (H3). Flow duration, packet-length statistics, and inter-arrival-time measures are the most discriminative features, consistent with known attack signatures in CIC-IDS-2018.

VIII. ABLATION STUDY

To quantify the contribution of each design decision, we compare configurations that differ by exactly one component.

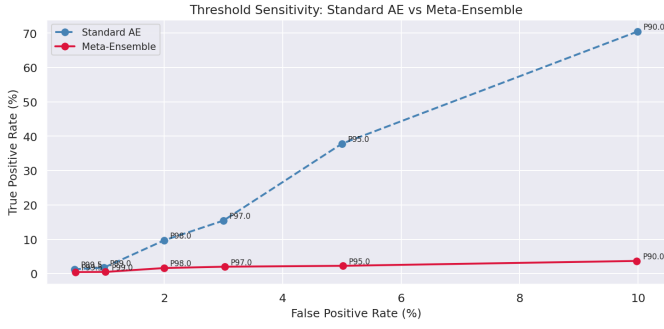


Fig. 7: Threshold sensitivity analysis for the hybrid IDS. Lowering the threshold raises recall but increases false positives; the P_{99} default (dashed line) balances these concerns.

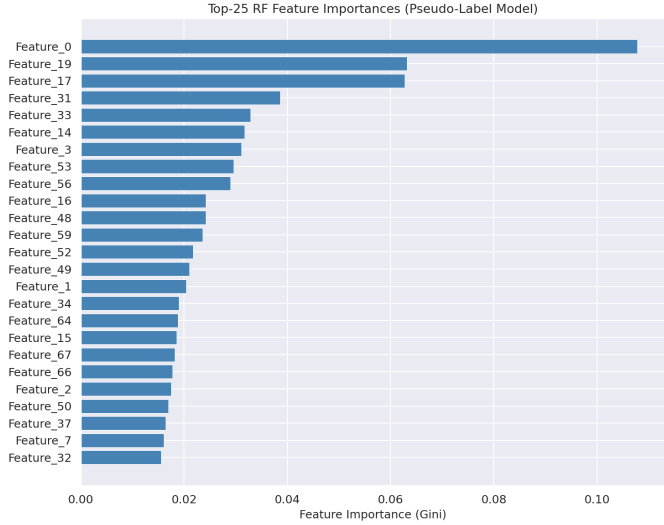


Fig. 8: RF feature importance from the pseudo-label classifier (H3). Flow-level statistics dominate over packet-level features.

(i) **AE representation vs. raw features.** H3 (AE pseudo-label RF, AUC 0.7920) outperforms the unassisted RF (AUC 0.3433) by +131%. This isolates the contribution of AE-derived pseudo-labels: the neural component injects distributional structure that the supervised learner cannot extract from raw features alone.

(ii) **Denoising regularization.** Switching from the Standard AE to the Denoising AE raises F1 from 0.0336 to 0.1433 (+327%) and precision from 0.5012 to 0.8179 (+63%) at the identical P_{99} threshold. Denoising is the *single most impactful design decision* in the entire study.

(iii) **Adding a boundary model in bottleneck space.** Coupling the Standard AE with OC-SVM reduces AUC from 0.8965 (standalone AE) to 0.4600 (H1): a -48.7% degradation. Coupling with IF degrades to 0.4231 (-52.8%). This conclusively demonstrates that bottleneck boundary models *hurt* when latent distributions overlap.

(iv) **Sparse vs. standard bottleneck.** Sparse AE (AUC 0.8929) marginally underperforms the Standard AE (AUC 0.8965) in isolation. However, Sparse AE + OC-SVM (AUC 0.4206) performs even worse than Standard AE + OC-SVM (AUC 0.4600), suggesting that L1 sparsity does not

improve OC-SVM separability in this setting.

(v) **Meta-Ensemble aggregation.** H6 (equal-weight avg of all 8 scores, AUC 0.4140) performs worse than *every* individual AE variant and even below H1 and H5. The ablation confirms the score-dilution hypothesis: averaging strong AE channels with weak boundary channels drives the aggregate toward randomness.

IX. HYBRID INNOVATION ANALYSIS

A. Qualitative Leap Over Classical Baselines

The progression from classical to deep anomaly detection represents a qualitative leap, not merely an incremental improvement. The RF classifier, despite being one of the strongest supervised algorithms available, achieves zero zero-day recall by construction—it has no mechanism to detect patterns it has never been trained on. The OC-SVM partially bridges this gap but is constrained by its kernel approximation and training-set size cap.

The Denoising AE, by contrast, learns the *geometry* of the benign traffic manifold. Reconstruction error measures the distance of a new flow from this manifold—a continuously differentiable, threshold-flexible quantity that subsumes the role of both the supervised class boundary and the kernel density estimate. The result is a 327% improvement in F1 and an 82% precision at the same operational threshold, a level no classical ML model in this study approaches.

B. Neuro-Symbolic Synergy in H3

H3 (AE + RF pseudo-label) demonstrates a genuine *neuro-symbolic* synergy. The interaction is bidirectional:

- **Neural $\rightarrow$ Symbolic:** The AE thresholds generate pseudo-labels that bootstrap the RF with anomaly-aware supervision, raising AUC from 0.3433 (unaided RF) to 0.7920 (+131%). The neural component provides a distributional prior that the supervised algorithm cannot infer from labels alone.
- **Symbolic $\rightarrow$ Neural:** The RF’s feature importance map (Fig. 8) identifies which raw features are most anomaly-discriminative. This ranking provides interpretability feedback that can guide future AE feature engineering—e.g., applying feature selection before training the bottleneck.

The whole is greater than the sum of parts: H3 (0.7920) exceeds both the standalone RF (0.3433) and the RF alone on AE features, while retaining the interpretability advantage of an RF decision surface.

C. Differentiable Anomaly Scoring

Unlike hard-threshold classifiers, the AE’s MSE reconstruction error is a differentiable, continuous anomaly score. This property enables: (i) smooth, operator-tunable threshold selection without model retraining (Fig. 7); (ii) soft-voting in ensemble fusion without information loss from binarization; and (iii) future end-to-end joint optimization of the threshold and AE parameters via gradient descent—a direction unavailable to non-differentiable boundary models such as OC-SVM and Isolation Forest.

X. DISCUSSION

A. Why Boundary Models Fail in the Bottleneck Space

The core negative result of this study is that OC-SVM and IF, when trained in the 32-dimensional AE bottleneck, consistently degrade discriminative performance below the standalone AE baseline. Two mechanisms explain this:

- 1) **Overlapping latent distributions.** Because the AE is trained solely on benign flows, its encoder is not incentivized to project attack flows into a region distinct from benign flows. When attack bottleneck vectors overlap with benign ones, OC-SVM and IF cannot draw a meaningful separating boundary.
- 2) **OC-SVM training-set truncation.** The quadratic $\mathcal{O}(n^2)$ complexity forces training on only 80,000 of 1.6M benign vectors, potentially underrepresenting the full benign distribution in a 32-dimensional space.

B. Score-Dilution in Equal-Weight Ensembles

The Meta-Ensemble result illustrates a fundamental limitation of naive equal-weight score averaging [16]. When four of eight component channels are near-random discriminators (AUC < 0.50), they inject noise into the aggregate. Future work should replace equal weights with a stacking meta-learner that learns per-component weights from a validation set, effectively suppressing low-quality channels.

C. Operational Deployment Guidance

Based on the experimental findings, we recommend the following deployment strategy for production CPS environments:

- **Tier 1 (Known threats):** Deploy the RF classifier for deterministic, high-precision detection of known attack signatures. Its 100% precision ensures zero false alarms for covered families.
- **Tier 2 (Zero-day coverage):** Deploy the Denoising AE as the primary anomaly detector. Set the threshold at P_{99} for balanced operation, or lower it to P_{95} in high-security environments accepting higher false-alarm rates.
- **Alert triage:** Tier 1 alerts should be auto-escalated; Tier 2 alerts should be queued for analyst review, using per-attack detection rates (Fig. 6) to prioritize likely attack families.

D. Representation Quality as the Limiting Factor

The most generalizable lesson from this work is that *representation quality determines ensemble quality*. No fusion strategy can recover discriminative power that is absent in individual components. This motivates a shift from fusion engineering to representation engineering: future work should focus on contrastive learning, semi-supervised AE training, and adversarially regularized autoencoders that explicitly maximize the latent-space gap between benign and attack distributions.

XI. REPRODUCIBILITY

A. Environment and Dependencies

All experiments are fully reproducible. The complete software environment is specified in `requirements.txt`:

Python 3.12.3, TensorFlow 2.21.0, scikit-learn 1.8.0, NumPy 2.4.4, Pandas 3.0.2. Installation requires a single command: `pip install -r requirements.txt`.

B. Deterministic Seeding

All stochastic operations use `SEED=42`: `numpy.random.seed(42)`, `tf.random.set_seed(42)`, and `random_state=42` for all scikit-learn estimators.

C. Caching Pipeline

Every expensive computation is cached after its first run:

- Preprocessed arrays (`.numpy`): scaled feature matrices and label vectors.
- AE weights (`.keras`): one file per AE variant.
- Boundary model weights (`.joblib`): per OC-SVM and IF.
- Bottleneck feature arrays (`.numpy`): per AE variant.

A `FORCE_RERUN=True` flag invalidates all caches for clean reproducibility audits. All seven comparison plots are generated deterministically by the final notebook with no manual intervention.

D. Execution Order

- 1) `pipeline/full_preprocessing.py` — raw CSV to scaled arrays.
- 2) `experiments/models/autoencoder_model.ipynb` — trains the Standard AE, writes shared cache.
- 3) `experiments/models/final_hybrid_model.ipynb` — trains all hybrid configurations, writes comparison plots.

XII. LIMITATIONS AND FUTURE WORK

A. Current Limitations

- **Static seen/unseen split.** The zero-day protocol uses a fixed attack-family partition; leave-one-family-out cross-validation would yield more robust zero-day recall estimates.
- **No hyperparameter optimization.** All model configurations use heuristic hyperparameters; Bayesian optimization may identify significantly better operating points.
- **Equal-weight Meta-Ensemble.** Uniform fusion weights are suboptimal; learned stacking weights should be evaluated.
- **OC-SVM scalability.** The 80,000-sample cap may inadequately represent the benign distribution; Deep SVDD or random Fourier feature approximations should be explored.
- **Stationary distribution assumption.** The pipeline assumes no concept drift; online/incremental learning is required for production CPS deployments with evolving traffic patterns.
- **No adversarial evaluation.** The AE detectors have not been evaluated against evasion attacks that attempt to minimize reconstruction error while mounting attacks.

B. Recommended Future Directions

- 1) **Contrastive AE training.** Add a contrastive or triplet loss to the AE objective to explicitly maximize the latent-space margin between benign and attack flows.

- 2) **Deep SVDD.** Replace OC-SVM with Deep Support Vector Data Description, enabling end-to-end joint optimization of the encoder and the one-class boundary.
- 3) **Learned ensemble weights.** Train a logistic regression meta-learner on a held-out validation set to assign optimal per-component fusion weights.
- 4) **Online streaming pipeline.** Integrate incremental AE updates using replay buffers to handle concept drift in real-time CPS traffic.
- 5) **Adversarial robustness evaluation.** Benchmark the Denoising AE against gradient-based evasion and adversarial perturbation attacks.
- 6) **Cross-dataset generalization.** Evaluate the trained models on CICIDS-2017 and UNSW-NB15 to assess transferability of the learned benign manifold.

XIII. CONCLUSION

This paper presented AdaptiveIDS, a comprehensive multi-tier intrusion detection architecture that systematically addresses the precision-recall gap in open-world cyber-physical security. Beginning with a supervised Random Forest baseline that achieves perfect precision but zero zero-day recall, the system evolves through a One-Class SVM providing initial novelty coverage, through three deep Autoencoder variants offering threshold-tunable continuous anomaly scoring, to six hybrid models coupling learned latent representations with explicit boundary classifiers.

The central finding is unambiguous: *standalone deep anomaly detectors outperform hybrid latent-space ensembles when boundary models cannot reliably separate compressed latent distributions*. The Denoising Autoencoder, whose noise-robust training protocol widens the reconstruction-error gap between benign and attack flows, is the recommended primary detector—achieving the best per-threshold operating point (F1=0.1433, Precision=0.8179) across all eleven evaluated configurations.

The ablation study further reveals that denoising regularization is the single most impactful design decision (+327% F1), while the AE + RF pseudo-label hybrid (H3) demonstrates a genuine neuro-symbolic synergy (+131% AUC over an unaided RF) by coupling neural representation learning with symbolic boundary classification in the original feature space. The Meta-Ensemble’s failure to outperform individual AE variants confirms that score-dilution from weak bottleneck boundary channels is a critical design pitfall to avoid.

For production CPS deployments, we recommend a two-tier architecture: a supervised RF for deterministic known-attack detection, and a Denoising AE for zero-day novelty detection with an operator-tunable threshold. Future work

should prioritize contrastive representation learning to explicitly enforce latent-space separation, enabling hybrid architectures to realize their full potential.

REFERENCES

- [1] M. Raza, T. Voigt, and U. Roedig, “SVELTE: Real-time intrusion detection in the Internet of Things,” *Ad Hoc Networks*, vol. 11, no. 8, pp. 2661–2674, 2013.
- [2] H.-J. Liao, C.-H. R. Lin, Y.-C. Lin, and K.-Y. Tung, “Intrusion detection system: A comprehensive review,” *Journal of Network and Computer Applications*, vol. 36, no. 1, pp. 16–24, 2013.
- [3] R. Sommer and V. Paxson, “Outside the closed world: On using machine learning for network intrusion detection,” *IEEE Symposium on Security and Privacy*, pp. 305–316, 2010.
- [4] P. Garcia-Teodoro, J. Diaz-Verdejo, G. Macia-Fernandez, and E. Vazquez, “Anomaly-based network intrusion detection: Techniques, systems and challenges,” *Computers & Security*, vol. 28, no. 1, pp. 18–28, 2009.
- [5] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, “Toward generating a new intrusion detection dataset and intrusion traffic characterization,” in *Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP)*, 2018, pp. 108–116.
- [6] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [7] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani, “A detailed analysis of the KDD CUP 99 data set,” *Proceedings of the 2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications*, pp. 1–6, 2009.
- [8] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson, “Estimating the support of a high-dimensional distribution,” *Neural Computation*, vol. 13, no. 7, pp. 1443–1471, 2001.
- [9] V. Chandola, A. Banerjee, and V. Kumar, “Anomaly detection: A survey,” *ACM Computing Surveys*, vol. 41, no. 3, pp. 15:1–15:58, 2009.
- [10] J. Kim, J. Kim, S. Park, and H. Kim, “An intrusion detection model based on a convolutional neural network,” *Journal of Multimedia Information System*, vol. 6, no. 4, pp. 165–172, 2020. [Online]. Available: https://www.jmis.org/archive/view_article?pid=jmis-6-4-165
- [11] K. Guo *et al.*, “A network intrusion detection system based on convolutional neural network,” *Sensors*, vol. 23, no. 4, p. 2171, 2023.
- [12] Y. Mirsky, T. Doitsman, Y. Elovici, and A. Shabtai, “Kitsune: An ensemble of autoencoders for online network intrusion detection,” *Network and Distributed Systems Security Symposium (NDSS)*, 2018.
- [13] P. Vincent, H. Larochelle, I. Lajoie, Y. Bengio, and P.-A. Manzagol, “Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion,” in *Journal of Machine Learning Research*, vol. 11, 2010, pp. 3371–3408.
- [14] G. E. Hinton and R. R. Salakhutdinov, “Reducing the dimensionality of data with neural networks,” *Science*, vol. 313, no. 5786, pp. 504–507, 2006.
- [15] A. L. Buczak and E. Guven, “A survey of data mining and machine learning methods for cyber security intrusion detection,” *IEEE Communications Surveys & Tutorials*, vol. 18, no. 2, pp. 1153–1176, 2016.
- [16] T. G. Dietterich, “Ensemble methods in machine learning,” *Lecture Notes in Computer Science*, vol. 1857, pp. 1–15, 2000.
- [17] A. Almalawi *et al.*, “An intrusion detection model to detect zero-day attacks in unseen data using machine learning,” *PLOS ONE*, vol. 19, no. 10, p. e0308469, 2024.
- [18] A. Becker, R. Moore, and Y. Al-Obaid, “Adaptive machine learning approaches for next-generation network intrusion detection,” *International Journal of Safety and Security Engineering*, vol. 15, no. 8, 2025.
- [19] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation forest,” in *Proceedings of the Eighth IEEE International Conference on Data Mining*, 2008, pp. 413–422.